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ACCESSORY PORT AND ACCESSORY FOR VEHICLES

Abstract

A crossbar for a vehicle may include a latch mechanism that provides a latch force that is maintained across various mounting conditions and/or changes in the mounting conditions. In various implementations, the latch mechanism is provided with a spring assembly coupled to a pawl this actuatable to releasably secure the crossbar to the vehicle.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of priority to U.S. Provisional Patent Application No. 63/562,209, entitled, “Accessory Port And Accessory for Vehicles”, filed on Mar. 6, 2024, the disclosure of which is hereby incorporated herein in its entirety.

INTRODUCTION

[0002] Accessory ports and accessories are provided for vehicles.

SUMMARY

[0003] The present description relates generally to an accessory port and an accessory for a vehicle. The accessory port may include a mechanical connector on a roof or a truck bed of a vehicle, and the accessory may be a crossbar, which also may be referred to herein as a rack, a roof rack, or a roof rack mount.

[0004] According to aspects of the disclosure, a crossbar for a vehicle is provided, the crossbar including a pawl configured to interact with a striker bar of a mechanical connector on the vehicle to releasably secure the crossbar to the mechanical connector; and a spring assembly mechanically coupled to the pawl and configured to provide a latching force tolerance for the pawl to releasably secure the crossbar to the mechanical connector. The crossbar may also include one or more sealing features configured to form a seal with the mechanical connector when the crossbar is mounted to the mechanical connector. The latching force tolerance may maintain a latching force for the pawl over a range of configurations of the one or more sealing features.

[0005] The one or more sealing features may include an outer sealing feature configured to form a first seal between the mechanical connector and an outermost surface of the crossbar, and an inner sealing feature configured to form a second seal between the striker bar of the mechanical connector and another surface of the crossbar.

[0006] The spring assembly may include a spring and screw assembly.

[0007] The spring assembly may include a leaf spring. The pawl may include a first portion and a second portion separate from the first portion, and the leaf spring may couple the first portion of the pawl to the second portion of the pawl.

[0008] The crossbar may also include a locking structure and an additional spring that biases the locking structure into a locked position that holds the pawl in a latched configuration. The crossbar may also include a paddle for controlling a latch mechanism comprising the pawl, the spring assembly, the locking structure, and the additional spring. Moving the paddle when the pawl is in the latched configuration moves a protrusion on a structure that is mechanically coupled to the paddle into contact with the locking structure to push the locking structure away from the locked position, compressing the additional spring, until the pawl is released into an unlatched configuration. The spring assembly may be further configured to allow a latch comprising the pawl to be latched and unlatched using a constant force over a range of the latch.

[0009] According to other aspects of the disclosure, a vehicle is provided that includes a mechanical connector having a striker bar; and a crossbar that includes a latch mechanism. The latch mechanism may include a pawl configured to interact with the striker bar of the mechanical connector to releasably secure the crossbar to the mechanical connector, and a spring assembly mechanically coupled to the pawl and configured to provide a latching force tolerance for the pawl to releasably secure the crossbar to the mechanical connector.

[0010] The vehicle may also include the mechanical connector mounted to a roof or a truck bed of the vehicle. The crossbar further may also include one or more sealing features configured to form a seal with the mechanical connector when the crossbar is mounted to the mechanical connector, wherein the latching force tolerance maintains a latching force for the pawl over a range of configurations of the one or more sealing features.

[0011] The spring assembly may include a die spring and screw assembly.

[0012] The spring assembly may include a leaf spring. The pawl may include a first portion and a

second portion separate from the first portion, and the leaf spring may couple the first portion of the pawl to the second portion of the pawl. The latch mechanism may also include a locking structure and an additional spring that biases the locking structure into a locked position that holds the pawl in a latched configuration.

[0013] The crossbar may also include a paddle for controlling the latch mechanism that includes the pawl, the spring assembly, the locking structure, and the additional spring. Moving the paddle when the pawl is in the latched configuration moves a protrusion on a structure that is mechanically coupled to the paddle into contact with the locking structure to push the locking structure away from the locked position, compressing the additional spring, until the pawl is released into an unlatched configuration. The spring assembly may be further configured to allow a latch comprising the pawl to be latched and unlatched using a constant force over a range of the latch.

[0014] According to other aspects of the disclosure, a latch mechanism for a crossbar for a vehicle is provided, the latch mechanism including a pawl configured to interact with a striker bar of a mechanical connector on the vehicle to releasably secure the crossbar to the mechanical connector, and a spring assembly mechanically coupled to the pawl and configured to provide a latching force tolerance for the pawl to releasably secure the crossbar to the mechanical connector. The pawl may include a first portion and a second portion separate from the first portion, and wherein a leaf spring couples the first portion of the pawl to the second portion of the pawl.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] Certain features of the subject technology are set forth in the appended claims. However, for purpose of explanation, several embodiments of the subject technology are set forth in the following figures.

[0016] FIGS. 1A and 1B illustrate schematic side views of example implementations of a vehicle having crossbars mounted thereto in accordance with one or more implementations.

[0017] FIGS. 2A and 2B illustrate schematic perspective views of example implementations of a vehicle having crossbars mounted thereto in accordance with one or more implementations.

[0018] FIG. 3 illustrates a side view of an example crossbar in accordance with one or more implementations.

[0019] FIG. 4 illustrates a side view of another example crossbar in accordance with one or more implementations.

[0020] FIG. 5 illustrates a cross-sectional side view of an example implementation of a crossbar in accordance with one or more implementations.

[0021] FIG. 6 illustrates a cross-sectional side view of another example implementation of a crossbar in accordance with one or more implementations.

[0022] FIG. 7 illustrates a cross-sectional side view of yet another example implementation of a crossbar in accordance with one or more implementations.

DETAILED DESCRIPTION

[0023] The detailed description set forth below is intended as a description of various configurations of the subject technology and is not intended to represent the only configurations in which the subject technology can be practiced. The appended drawings are incorporated herein and constitute a part of the detailed description. The detailed description includes specific details for the purpose of providing a thorough understanding of the subject technology. However, the subject technology is not limited to the specific details set forth herein and can be practiced using one or more other implementations. In one or more implementations, structures and components are shown in block diagram form in order to avoid obscuring the concepts of the subject technology.

[0024] Aspects of the subject technology described herein relate to an accessory port and an

accessory for a vehicle, such as an electric vehicle.

[0025] FIG. 1A is a diagram illustrating an example implementation of an apparatus as described herein. In the example of FIG. 1A, the apparatus includes a moveable apparatus implemented as a vehicle **100**. As shown, the vehicle **100** may include one or more batteries **110**. The battery **110** may be coupled to an electrical system of the vehicle **100**, to receive power for charging of the battery and/or to provide power to one or more electronic components of the vehicle electrical system.

[0026] In one or more implementations, the vehicle **100** may be an electric vehicle having one or more electric motors that drive the wheels **102** of the vehicle using electric power from the battery **110**. The battery **110** may include one or more battery modules, which may include one or more battery cells, or may be provided without any battery modules (e.g., in a cell-to-pack configuration). In one or more implementations, the vehicle **100** may also, or alternatively, include one or more chemically powered engines, such as a gas-powered engine or a fuel cell powered motor. For example, electric vehicles can be fully electric or partially electric (e.g., hybrid or plug-in hybrid).

[0027] In the example of FIG. 1A, the vehicle **100** is implemented as a truck (e.g., a pickup truck) having a roof **122** and a truck bed **124**.

[0028] As shown, the vehicle **100** may include one or more respective mechanical connectors **159** (e.g., accessory ports). For example, the mechanical connectors **159** may be disposed on and/or embedded within the roof **122** of the vehicle **100**, and/or on a surface or a wall of the truck bed **124** of the vehicle **100**.

[0029] In the example of FIG. 1A, two mechanical connectors **159** are visible on the roof **122** of the vehicle **100** (e.g., at or near the left side of the vehicle that is visible in FIG. 1A). In one or more implementations, two additional mechanical connectors **159** may be symmetrically disposed on the roof **122** of the vehicle **100**, such as at or near the right side of the vehicle that is not visible in FIG. 1A. In such implementations, the roof **122** of the vehicle **100** may include four mechanical connectors **159**. As shown in FIG. 1A, in one or more implementations, the vehicle **100** may (e.g., in addition to and/or instead of one or more of the mechanical connectors **159** on the roof **122** of the vehicle) include one or more mechanical connectors **159** on or near a wall of the truck bed **124** of the vehicle.

[0030] As shown in FIG. 1A, one or more crossbars **120** may be mounted to the vehicle **100** using the mechanical connectors **159**. For example, each of the crossbars **120** may be removably mounted to one or more of the mechanical connectors **159**. For example, each crossbar **120** may be removably mounted to two mechanical connectors on opposing sides of the roof **122** or the truck bed **124**.

[0031] In the example of FIG. 1A, the crossbars **120** are mounted to the mechanical connectors **159** of a vehicle **100** implemented as a pickup truck. However, in one or more other use cases, the crossbars **120** and the mechanical connectors **159** may be mounted to another type of vehicle. For example, FIG. 1B illustrates another use case in which the vehicle **100** including the battery **110**, the mechanical connectors **159**, and the crossbars **120** is implemented as a sport utility vehicle (SUV), such as an electric sport utility vehicle. In the example of FIG. 1B, the vehicle **100** may include a cargo storage area in at least a rear portion of the vehicle that is enclosed within the vehicle **100** (e.g., behind a row of seats within a cabin of the vehicle). As shown, in this example, the vehicle **100** may include two mechanical connectors **159** located on a left side of the roof **122** of the SUV. One or more (e.g., two) additional mechanical connectors **159** may also be provided (e.g., symmetrically with the two mechanical connectors **159** on the left side of the roof **122** of the vehicle) on a right side of the roof **122** of the vehicle. In other implementations, the vehicle **100** may implemented as another type of electric truck, an electric delivery van, an electric automobile, an electric car, an electric motorcycle, an electric scooter, an electric passenger vehicle, an electric passenger or commercial truck, a hybrid vehicle, or other vehicles such as sea or air transport

vehicles, planes, helicopters, submarines, boats, or drones, and/or any other movable apparatus having mechanical connectors for crossbars.

[0032] FIG. 2A illustrates a perspective view a vehicle **100** implemented as a pickup truck as in FIG. 1A, and showing how the crossbars **120** may extend across the width of the vehicle and be mounted at or near opposing sides of the vehicle (e.g., mounted to the mechanical connectors **159** on the opposing sides of the vehicle). FIG. 2B illustrates a perspective view a vehicle **100** implemented as an SUV as in FIG. 1B, and showing how the crossbars **120** may extend across the width of the vehicle and be mounted at or near opposing sides to the vehicle (e.g., mounted to the mechanical connectors **159** on the opposing sides of the vehicle).

[0033] FIG. 3 illustrates a side view of an example crossbar **120** in accordance with one or more implementations. In the example of FIG. 3, the mechanical connector **159** may be (e.g., permanently) attached to the vehicle using bolts. In the example of FIG. 3, a paddle **300** of the crossbar **120** is shown in an open position. In the open position of FIG. 3, the paddle **300** may be pressed or pulled upward to operate a latch mechanism within the crossbar **120** to release the crossbar **120** from the mechanical connector, and/or pressed or pushed down operate the latch mechanism to latch or clamp the crossbar **120** to the mechanical connector **159**, as described in further detail hereinafter. As shown, a lock **302** may be provided under the paddle **300**. For example, the lock **302** may be switched from an unlocked position that allows the paddle **300** to be actuated up to release the crossbar **120** and/or down to latch the crossbar **120** to the mechanical connector **159**.

[0034] FIG. 4 illustrates a side view of another example crossbar in accordance with one or more implementations. In the example of FIG. 4, the mechanical connector **159** may include a mechanism **400** and a tab **402**. For example, the mechanism **400** may be a release mechanism, and the tab **402** may be mechanically coupled to the release mechanism to release the mechanical connector **159** from the vehicle **100**. In the example of FIG. 4, the paddle **300** is shown in a closed position. In the closed position of FIG. 4, the latching mechanism within the crossbar **120** clamps the crossbar **120** to the mechanical connector **159**.

[0035] FIGS. 5, 6, and 7 show internal features of various latch mechanisms that may be included within the crossbar **120**, and that may be actuated by the paddle **300** of FIGS. 3 and 4 to release and/or latch the crossbar **120** from and/or to the mechanical connector **159**.

[0036] FIG. 5 illustrates a cross-sectional side view of the crossbar of FIG. 3 in accordance with one or more implementations. In the example, of FIG. 5, the crossbar **120** is shown in the same orientation as in FIGS. 3 and 4, and with a portion of the housing of the crossbar removed to show the latching mechanism **501**. As shown, the crossbar **120** may include a latch mechanism **501** that is operable to releasably couple the crossbar to the mechanical connector **159** using a pawl **500** that interfaces with a striker bar **502** on the mechanical connector **159**. In the example of FIG. 5, the latching mechanism **501** is shown in an unlatched configuration. In this example, moving the paddle downward may lift a structure **503** that is coupled to the paddle **300** and the pawl **500**. The structure **503** may pull upward on one side of the pawl **500** until a locking structure **504** moves (e.g., snaps) into place to hold the pawl **500** in a latched position around the striker bar **502**. Once latched, lifting the paddle **300** upward may move a protrusion **506** downward to rotate the locking structure **504** to allow the pawl **500** to move (e.g., snap) back into the unlatched position shown in FIG. 5.

[0037] In the example of FIG. 5, the pawl **500** may provide a limited amount of tolerance with respect to the clamping force of the latch mechanism **501** to the mechanical connector **159**. For example, a sealing material **508** may be provided on the crossbar **120** to provide a sealed interface between the crossbar **120** and the mechanical connector **159** when the crossbar **120** is latched. The sealing material **508** may be a compliant material, that may change in thickness or compliance over time. In one or more implementations, one or more adjustment screws **510** may be provided for modifying the clamping force of the latching mechanism **501**. However, it may be desirable to be

able to provide a latching mechanism that incorporates an internal tolerance for the clamping force (e.g., to adjust for changes in thickness and/or compliance of the sealing material, without adjustments to the adjustment screws **510**).

[0038] FIG. **6** illustrates a cross-sectional side view of the crossbar of FIG. **4** in an implementation in which the crossbar includes a latch mechanism **601**. For example, the latch mechanism **601** may provide the crossbar **120** with additional clamping force tolerance, relative to that of the latch mechanism **501** of FIG. **5**. In the example, of FIG. **6**, the crossbar **120** is shown in the same orientation as in FIGS. **3** and **4**, and with a portion of the housing of the crossbar removed to show the latch mechanism **601**. As shown in FIG. **6**, the crossbar **120** may releasably couple to the mechanical connector **159** using a pawl **600** that interfaces with the striker bar **502** on the mechanical connector **159**.

[0039] In the example of FIG. **6**, the latch mechanism **601** is shown in a latched configuration with a locking structure **609** holding the pawl **600** in a latched position around the striker bar **502**. In this example, lifting the paddle **300** upward may move a protrusion **611** downward to rotate the locking structure **609** to allow the pawl **600** to rotate (e.g., snap) into an unlatched position. Once unlatched, moving the paddle **300** downward may lift a structure **607** that is coupled to the paddle **300** and/or the pawl **600**. Moving the structure **607** upward may cause or allow one side of the pawl **600** to move upward while the other side of the pawl moves downward until the locking structure **609** rotates (e.g., snaps) into place to hold the pawl **600** in the latched position of FIG. **6**.

[0040] In contrast with the implementation of FIG. **5**, the crossbar **120** of FIG. **6** may include a spring assembly, such as a die spring and screw assembly **604** (e.g., including a spring **605** and a screw **603**). For example, the die spring and screw assembly **604** may apply an upward force on the side of the pawl **600** that is away from the locking structure **609**, along some or all of the range of motion of the pawl **600**. In this way, the die spring and screw assembly **604** may provide benefits (e.g., relative to the implementation of FIG. **5**), such as additional tolerance accommodation and/or a constant force being applied to the latch (e.g., including the pawl **600**), such as over a range (e.g., a full range) of motion of the latch. In the example of FIG. **6**, the same force may be provided at the paddle **300** to unlatch and latch the crossbar **120** to the mechanical connector **159** (e.g., due to the over center design of the system in FIG. **6**). In the example of FIG. **6**, a die spring and screw assembly **604** is used to, for example, provide a latching force tolerance for the pawl to releasably secure the crossbar to the mechanical connector. However, other configurations of latch assemblies are also possible, including using other spring assemblies in the latch mechanism.

[0041] For example, FIG. **7** illustrates a cross-sectional side view of another example implementation of the crossbar of FIG. **4**, in an implementation in which the crossbar includes a latch mechanism **701** with another spring assembly **704**. For example, the latch mechanism **701** may also provide the crossbar **120** with additional clamping force tolerance, relative to that of the latch mechanism **501** of FIG. **5**. In the example, of FIG. **7**, the crossbar **120** is shown in an orientation opposite to the orientation shown in FIGS. **3** and **4**. Although the striker bar **502** is not shown in FIG. **7**, the crossbar **120** may releasably couple to the mechanical connector **159** using a pawl **700** that interfaces with the striker bar **502** on the mechanical connector **159**.

[0042] In the example of FIG. **7**, the latch mechanism **701** is shown in a latched configuration with a locking structure **709** holding the pawl **700** in a latched position (e.g., with a curved portion of the pawl **700** extending partially around the striker bar **502**). As shown, an additional spring **713** may be provided in the latch mechanism **701** that biases the locking structure **709** toward the locked position shown in FIG. **7**. In this example, lifting the paddle **300** upward may move a protrusion **711** downward to rotate the locking structure **709** (e.g., pushing against the biasing force of the additional spring **713**) to allow the pawl **700** to move (e.g., snap) into an unlatched position. Once unlatched, moving the paddle **300** downward may lift a structure **707** (e.g., a structure **707** having the protrusion) that is coupled to the paddle **300** and/or the pawl **700**. Moving the structure **707** upward may cause or allow a first portion **700A** of the pawl **700** to move upward, which may rotate

a second portion **700B** of the pawl (e.g., around the striker bar **502**) until the locking structure **709** rotates (e.g., snaps) into place to hold the first portion **700A** of the pawl **700** in the latched position of FIG. 7.

[0043] As shown, the spring assembly **704** of FIG. 7 includes a leaf spring **705**. As shown, the leaf spring **705** may couple a first portion **700A** of the pawl **700** to the second portion **700B** of the pawl **700**. For example, the first portion **700A** of the pawl **700** and the second portion **700B** of the pawl **700** may be separated by a gap **706**. The leaf spring **705** may span the gap **706**, with a first end of the leaf spring **705** attached to first portion **700A** of the pawl **700**, and a second end of the leaf spring **705** attached to the second portion **700B** of the pawl **700**. In the latched configuration of FIG. 7, the leaf spring **705** may be flexed into a substantially flat shape. When the first portion **700A** of the pawl **700** is released by the locking structure **709**, the leaf spring **705** may contract into a curved (e.g., a wide U-shape), pulling the first portion **700A** and the second portion **700B** of the pawl **700** toward each other. In the flexed configuration shown in FIG. 7, the leaf spring **705** may exert a force on the second portion **700B** of the pawl **700** that holds the second portion **700B** against the striker bar of the mechanical connector **159**, even if the relative positions of the pawl **700** and the striker bar vary (e.g., up to a range of flexure of the leaf spring).

[0044] In this way, the leaf spring **705** may provide a spring-loaded compliance in the pawl **700** that helps to provide a desired latching force even in different configurations of a sealing feature **708**. For example, the leaf spring **705** may allow manufacturing variations in the sealing feature **708**, and/or changes in compliance and/or thickness of the sealing feature **708** over time (e.g., due to wear) to be tolerated by the latch mechanism **701**, without substantially changing the latching force with which the latch mechanism **701** latches the crossbar **120** to a vehicle (e.g., to a striker bar on the mechanical connector **159**). For example, if the sealing feature **708** becomes more compressed over time, causing the striker bar to move away from the latching portion of the second portion **700B** of the pawl **700**, the leaf spring **705** may provide a force that compensates for that relative motion to maintain the latching force provided by the pawl **700**.

[0045] As illustrated by FIGS. 1A-7, in one or more implementations, a crossbar **120** for a vehicle **100** may be provided. The crossbar **120** may include a pawl (e.g., pawl **600** or pawl **700**) configured to interact with a striker bar **502** of a mechanical connector **159** on the vehicle **100** to releasably secure the crossbar **120** to the mechanical connector **159**. The crossbar **120** may also include a spring assembly (e.g., die spring and screw assembly **604** of FIG. 6 or leaf spring **705** of FIG. 7) mechanically coupled to the pawl (e.g., to provide a spring force on a portion of the pawl). The crossbar **120** may also include one or more sealing features (e.g., sealing material **508**, sealing features **606** and/or **608**, or sealing features **708**) configured to form a seal with the mechanical connector **159** when the crossbar **120** is mounted to the mechanical connector **159**. For example, the one or more sealing features may include an outer sealing feature **606** (e.g., configured to form a first seal between the mechanical connector **159** and an outermost surface of the crossbar **120**). The one or more sealing features may also include an inner sealing feature **608** (e.g., configured to form a second seal between the striker bar **502** of the mechanical connector **159** and another surface of the crossbar **120**). As discussed herein, the spring assembly may be configured to provide a tolerance accommodation for mounting the crossbar **120** to the mechanical connector **159**, and/or to allow a latch comprising the pawl to be latched and unlatched using a constant force over a range (e.g., a full range) of the latch.

[0046] In one or more implementations, a vehicle **100** may be provided, the vehicle **100** including a mechanical connector **159** having a striker bar **502**. The vehicle **100** may also include a crossbar **120**. The crossbar **120** may include a pawl (e.g., pawl **600** or pawl **700**) configured to interact with the striker bar **502** of the mechanical connector **159** to releasably secure the crossbar **120** to the mechanical connector **159**. The crossbar **120** may also include a spring assembly (e.g., die spring and screw assembly **604** of FIG. 6 or leaf spring **705** of FIG. 7) mechanically coupled to the pawl.

[0047] In one or more implementations, the latching force tolerance maintains a latching force for

the pawl over a range of configurations of the one or more sealing features (e.g., due to mechanical variations in the sealing features and/or wear of the sealing features over time). In one or more implementations, the spring assembly may include a leaf spring **705**. For example, the pawl may include a first portion **700A** and a second portion **700B** separate from the first portion, and the leaf spring **705** may couple the first portion **700A** of the pawl to the second portion **700B** of the pawl. In one or more implementations, the crossbar (e.g., the latching mechanism of the crossbar) may also include a locking structure **709** and an additional spring **713** that biases the locking structure **709** into a locked position that holds the pawl in a latched configuration (e.g., as shown in FIG. 7). In one or more implementations, the crossbar may also include a paddle **300** for controlling a latch mechanism **701** that includes the pawl, the spring assembly (e.g., spring assembly **704**), the locking structure **709**, and the additional spring **713**. Moving the paddle when the pawl is in the latched configuration may move a protrusion **711** on a structure **707** that is mechanically coupled to the paddle into contact with the locking structure **709** to push the locking structure **709** away from the locked position, compressing the additional spring **713**, until the pawl **700** is released into an unlatched configuration. In one or more implementations, the spring assembly is further configured to allow a latch comprising the pawl to be latched and unlatched using a constant force over a range of the latch.

[0048] As used herein, the phrase “at least one of” preceding a series of items, with the term “and” or “or” to separate any of the items, modifies the list as a whole, rather than each member of the list (i.e., each item). The phrase “at least one of” does not require selection of at least one of each item listed; rather, the phrase allows a meaning that includes at least one of any one of the items, and/or at least one of any combination of the items, and/or at least one of each of the items. By way of example, the phrases “at least one of A, B, and C” or “at least one of A, B, or C” each refer to only A, only B, or only C; any combination of A, B, and C; and/or at least one of each of A, B, and C.

[0049] The predicate words “configured to”, “operable to”, and “programmed to” do not imply any particular tangible or intangible modification of a subject, but, rather, are intended to be used interchangeably. In one or more implementations, a processor configured to monitor and control an operation or a component may also mean the processor being programmed to monitor and control the operation or the processor being operable to monitor and control the operation. Likewise, a processor configured to execute code can be construed as a processor programmed to execute code or operable to execute code.

[0050] Phrases such as an aspect, the aspect, another aspect, some aspects, one or more aspects, an implementation, the implementation, another implementation, some implementations, one or more implementations, an embodiment, the embodiment, another embodiment, some implementations, one or more implementations, a configuration, the configuration, another configuration, some configurations, one or more configurations, the subject technology, the disclosure, the present disclosure, other variations thereof and alike are for convenience and do not imply that a disclosure relating to such phrase(s) is essential to the subject technology or that such disclosure applies to all configurations of the subject technology. A disclosure relating to such phrase(s) may apply to all configurations, or one or more configurations. A disclosure relating to such phrase(s) may provide one or more examples. A phrase such as an aspect or some aspects may refer to one or more aspects and vice versa, and this applies similarly to other foregoing phrases.

[0051] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration”. Any embodiment described herein as “exemplary” or as an “example” is not necessarily to be construed as preferred or advantageous over other implementations. Furthermore, to the extent that the term “include”, “have”, or the like is used in the description or the claims, such term is intended to be inclusive in a manner similar to the term “comprise” as “comprise” is interpreted when employed as a transitional word in a claim.

[0052] All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the

art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. No claim element is to be construed under the provisions of 35 U.S.C. § 112(f) unless the element is expressly recited using the phrase “means for” or, in the case of a method claim, the element is recited using the phrase “step for”. [0053] The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not intended to be limited to the aspects shown herein, but are to be accorded the full scope consistent with the language claims, wherein reference to an element in the singular is not intended to mean “one and only one” unless specifically so stated, but rather “one or more”. Unless specifically stated otherwise, the term “some” refers to one or more. Pronouns in the masculine (e.g., his) include the feminine and neuter gender (e.g., her and its) and vice versa. Headings and subheadings, if any, are used for convenience only and do not limit the subject disclosure.

[0054] The title, background, brief description of the drawings, abstract, and drawings are hereby incorporated into the disclosure and are provided as illustrative examples of the disclosure, not as restrictive descriptions. It is submitted with the understanding that they will not be used to limit the scope or meaning of the claims. In addition, in the detailed description, it can be seen that the description provides illustrative examples and the various features are grouped together in various implementations for the purpose of streamlining the disclosure. The method of disclosure is not to be interpreted as reflecting an intention that the claimed subject matter requires more features than are expressly recited in each claim. Rather, as the claims reflect, inventive subject matter lies in less than all features of a single disclosed configuration or operation. The claims are hereby incorporated into the detailed description, with each claim standing on its own as a separately claimed subject matter.

[0055] The claims are not intended to be limited to the aspects described herein, but are to be accorded the full scope consistent with the language of the claims and to encompass all legal equivalents. Notwithstanding, none of the claims are intended to embrace subject matter that fails to satisfy the requirements of the applicable patent law, nor should they be interpreted in such a way.

Claims

1. A crossbar for a vehicle, the crossbar comprising: a pawl configured to interact with a striker bar of a mechanical connector on the vehicle to releasably secure the crossbar to the mechanical connector; and a spring assembly mechanically coupled to the pawl and configured to provide a latching force tolerance for the pawl to releasably secure the crossbar to the mechanical connector.
2. The crossbar of claim 1, further comprising one or more sealing features configured to form a seal with the mechanical connector when the crossbar is mounted to the mechanical connector, wherein the latching force tolerance maintains a latching force for the pawl over a range of configurations of the one or more sealing features.
3. The crossbar of claim 2, wherein one or more sealing features comprise: an outer sealing feature configured to form a first seal between the mechanical connector and an outermost surface of the crossbar; and an inner sealing feature configured to form a second seal between the striker bar of the mechanical connector and another surface of the crossbar.
4. The crossbar of claim 1, wherein the spring assembly comprises a spring and screw assembly.
5. The crossbar of claim 1, wherein the spring assembly comprises a leaf spring.
6. The crossbar of claim 5, wherein the pawl comprises a first portion and a second portion separate from the first portion, and wherein the leaf spring couples the first portion of the pawl to the second

portion of the pawl.

7. The crossbar of claim 1, further comprising: a locking structure; and an additional spring that biases the locking structure into a locked position that holds the pawl in a latched configuration.

8. The crossbar of claim 7, further comprising a paddle for controlling a latch mechanism comprising the pawl, the spring assembly, the locking structure, and the additional spring, wherein moving the paddle when the pawl is in the latched configuration moves a protrusion on a structure that is mechanically coupled to the paddle into contact with the locking structure to push the locking structure away from the locked position, compressing the additional spring, until the pawl is released into an unlatched configuration.

9. The crossbar of claim 1, wherein the spring assembly is further configured to allow a latch comprising the pawl to be latched and unlatched using a constant force over a range of the latch.

10. A vehicle, comprising: a mechanical connector having a striker bar; and a crossbar comprising a latch mechanism comprising: a pawl configured to interact with the striker bar of the mechanical connector to releasably secure the crossbar to the mechanical connector; and a spring assembly mechanically coupled to the pawl and configured to provide a latching force tolerance for the pawl to releasably secure the crossbar to the mechanical connector.

11. The vehicle of claim 10, wherein the mechanical connector is mounted to a roof or a truck bed of the vehicle.

12. The vehicle of claim 10, the crossbar further comprising one or more sealing features configured to form a seal with the mechanical connector when the crossbar is mounted to the mechanical connector, wherein the latching force tolerance maintains a latching force for the pawl over a range of configurations of the one or more sealing features.

13. The vehicle of claim 10, wherein the spring assembly comprises a die spring and screw assembly.

14. The vehicle of claim 10, wherein the spring assembly comprises a leaf spring.

15. The vehicle of claim 14, wherein the pawl comprises a first portion and a second portion separate from the first portion, and wherein the leaf spring couples the first portion of the pawl to the second portion of the pawl.

16. The vehicle of claim 10, the latch mechanism further comprising: a locking structure; and an additional spring that biases the locking structure into a locked position that holds the pawl in a latched configuration.

17. The vehicle of claim 16, the crossbar further comprising a paddle for controlling the latch mechanism comprising the pawl, the spring assembly, the locking structure, and the additional spring, wherein moving the paddle when the pawl is in the latched configuration moves a protrusion on a structure that is mechanically coupled to the paddle into contact with the locking structure to push the locking structure away from the locked position, compressing the additional spring, until the pawl is released into an unlatched configuration.

18. The vehicle of claim 1, wherein the spring assembly is further configured to allow a latch comprising the pawl to be latched and unlatched using a constant force over a range of the latch.

19. A latching mechanism for a crossbar for a vehicle, the latching mechanism comprising: a pawl configured to interact with a striker bar of a mechanical connector on the vehicle to releasably secure the crossbar to the mechanical connector; and a spring assembly mechanically coupled to the pawl and configured to provide a latching force tolerance for the pawl to releasably secure the crossbar to the mechanical connector.

20. The latching mechanism of claim 19, wherein the pawl comprises a first portion and a second portion separate from the first portion, and wherein a leaf spring couples the first portion of the pawl to the second portion of the pawl.
